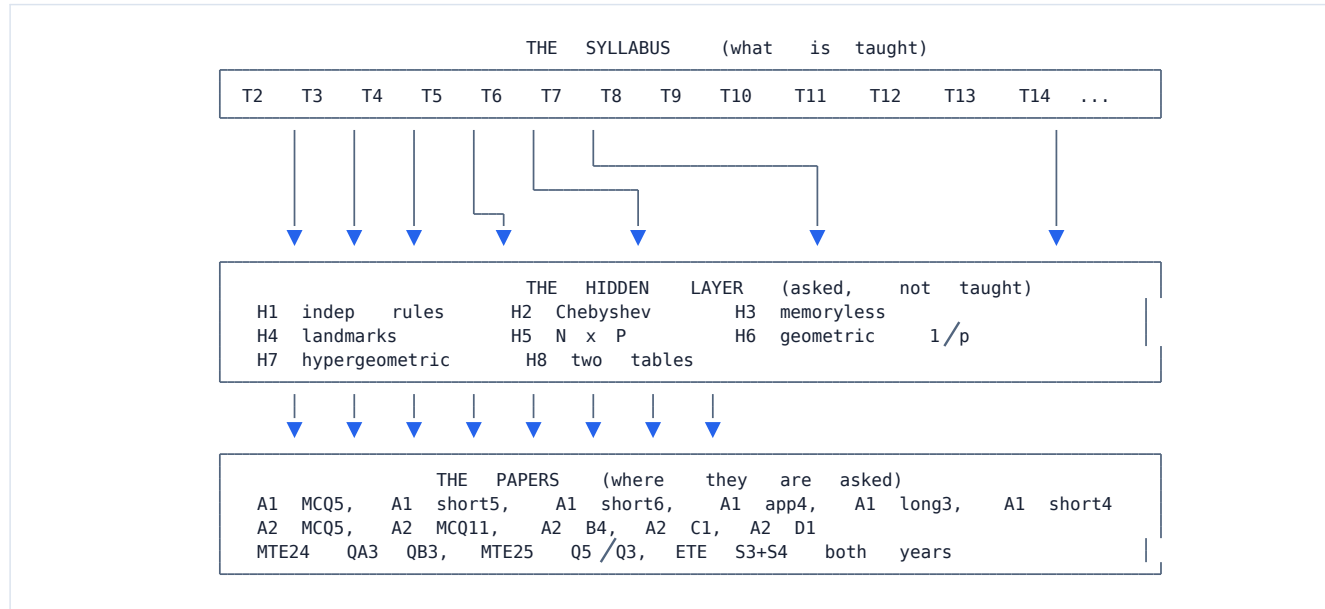
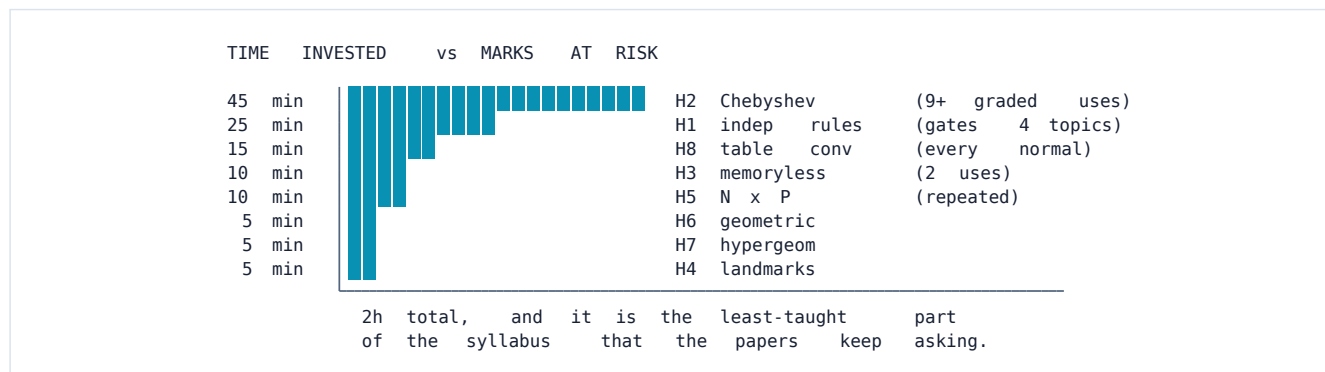


11 BEYOND SLIDES: what is asked but never taught

THE HIDDEN LAYER, CONNECTED TO THE COURSE



THE COST-BENEFIT OF DRILLING THE HIDDEN LAYER



THE INDEPENDENCE RULES DRAWN (H1, the highest-value)

INDEPENDENT X, Y:		
$E(XY)$	$= E(X) \cdot E(Y)$	needs INDEPENDENCE
$\text{Var}(X+Y)$	$= \text{Var}(X) + \text{Var}(Y)$	needs INDEPENDENCE
$\text{Var}(X-Y)$	$= \text{Var}(X) + \text{Var}(Y)$	the minus STAYS PLUS
$E(X+Y)$	$= E(X) + E(Y)$	ALWAYS, no condition
downstream uses:		
$\text{Var}(\text{sum of two dice}) = 35/12 + 35/12 = 35/6$		
↳ feeds the Chebyshev 35/54 bound		
$\text{Var}(\bar{X}) = \sigma^2/n$	$=$ this rule applied n times	
↳ feeds the CLT standard error		
$\text{Var}(T)$ for estimator comparison		
↳ the 6-8 mark estimation question		
Poisson additivity: $\text{Poi}(\lambda_1) + \text{Poi}(\lambda_2) = \text{Poi}(\lambda_1 + \lambda_2)$		
without H1, HALF the estimation block is unreachable.		

These eight items are asked in the held graded material but are missing, lightly treated, or easy to overlook in the main converted teaching decks. They are high-value review items and several are prerequisites for later topics.

H1. Independence rules of random variables

why hidden	the notes deck teaches expectation to p094 and variance from p095, but the
product rule and the variance-of-a-sum rule sit on edges the conversion did not	
capture as a named slide. Yet they gate four downstream topics.	
what it is	$E(XY) = E(X) E(Y)$ when X, Y independent
$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ when independent	
$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$ when independent	(the minus stays plus)
contrast: $E(X + Y) = E(X) + E(Y)$ needs NO independence	
why it matters	every dice-sum variance, every $\text{Var}(\bar{X})$, every estimator variance, every
Poisson additivity uses it. Without it, half the estimation block cannot be done.	
drill	$\text{Var}(\text{one die}) = 35/12 \rightarrow \text{Var}(\text{sum of two}) = 35/6 \rightarrow$ the 35/54 Chebyshev bound
$\text{Var}(\bar{X}) = \sigma^2/n$ is this rule applied n times	
real ask	R25S3-A1 asks $E[XY] = E[X]E[Y]$ directly (confirmed this is a graded question)

H2. Chebyshev's inequality (whole topic)

why hidden	batch 1 had no Chebyshev slide at all. The L10-11 deck arrived 15 Sep and closed
the gap, but it is the ONE deck that is genuinely new; everything else re-teaches.	
what it is	$P(X-\mu \geq k \sigma) \leq 1/k^2$, complement form $\geq 1 - 1/k^2$
k in sigma units. $k=2 \rightarrow 3/4$, $k=3 \rightarrow 8/9$, $k=4 \rightarrow 15/16$.	
No distribution assumed. Always an upper bound on the tail.	
why it matters	graded 9+ times: A1 MCQ5, MCQ10, short5, short6, app4; MTE 2024 QA3, QB3;
MTE 2025 Q5; ETE S3 and S4 both years; 2025-26 bundle Q5.	
drill	dice 35/54 vs exact 1/3; 600 throws 19/24; marks 75 percent; inverse c from a
stated bound (solve $1/k^2 =$ the given bound)	
trap	Q3 of the L10-11 deck mis-states its own row (errata 15), use Q1 and Q2 only

H3. The memoryless property

why hidden	no slide states $P(T > s+t \mid T > s) = P(T > t)$.
what it is	the exponential does not age: surviving s more, given you already survived,
equals surviving s from scratch.	
why it matters	A2 MCQ Q5 and A2 section C Q1 both use it. It is one line to state and a
full mark in a short answer.	
drill	$P(T > 5 \mid T > 3) = P(T > 2) = e^{(-2\lambda)}$.
This equals $e^{-1} = 0.3679$ only when $\lambda=0.5$.	

H4. The 68 / 95 / 99.7 landmarks

why hidden	no slide prints the empirical rule.
what it is	for a normal distribution: 68.27 percent within 1 sigma,
95.45 within 2, 99.73 within 3.	
why it matters	A2 MCQ Q11 uses it. Also the fast sanity check for any normal answer.
drill	"what fraction lies within 2 sigma" $\rightarrow$ 95.45.

H5. Expected count, $N \times P$

why hidden	slides compute probabilities and expectations separately, never the "out of N
items, how many" phrasing as a named method.	
what it is	expected number of items = $N \times P(\text{success})$. It is expectation applied to a count.
why it matters	A1 app, A2 B4 (5000 batteries), A2 D1 (10000 bulbs), the irregular die count
(0.5499), and the new E25SUM-Q13 (800 families) and R25S4-B1 (heights > 6ft).	
drill	out of 10000 sets, expected = $10000 \times (3/8)^{10} = 0.5499$
trap	report $N \times P$, not P . 0.5499, not $5.5e-5$.

H6. Geometric distribution

why hidden	the notes deck uses a geometric count (p057-059) but never names $E = 1/p$ as a
result.	
what it is	X = trials until the first success. $E(X) = 1/p$, $\text{Var}(X) = q/p^2$.
why it matters	A1 short Q4 asks the geometric expectation, $E = 2$. Graded once in batch 1,
so it is live.	
drill	$E = 1/p$. For $p = 1/2$, $E = 2$. The "number of failures before" convention gives
$1/p - 1$, read the wording.	

H7. Hypergeometric mean

why hidden	no slide. It appears only in a question.
what it is	drawing WITHOUT replacement from a finite lot: mean = $n \times K/N$, the same value as
the binomial mean at the same p .	
why it matters	A1 long Q3: lot of 25, 5 defective, sample of 4, mean = $4 \times 5/25 = 0.80$
exactly, both with and without replacement.	
drill	nK/N , and note it equals np here. (errata 11: the key prints 0.808, wrong.)

H8. The two normal table conventions

why hidden	the deck uses a table but does not name $\phi(z)$ vs $F(z)$ as two different
conventions.	
what it is	$\phi(z)$ is often tabulated as the area from 0 to z ; $F(z)$ is the cumulative from
$-\infty$ to z . They differ by 0.5 for $z > 0$.	
why it matters	every normal and CLT numeric goes through a table. Mixing the conventions is
the single biggest source of normal marks lost.	
drill	$P(Z < 0.12)$: from ϕ table, $0.5 + 0.0478 = 0.5478$. From F table, read 0.5478
directly. Same answer, different route. Always decide which table the question	
gives BEFORE substituting.	

The one-block summary (all eight, one line each)

H1	$E(XY)=E(X)E(Y)$, $\text{Var}(X+Y)=\text{Var}(X)+\text{Var}(Y)$, $\text{Var}(X-Y)$ adds	independence rules
H2	Chebyshev, k in σ , $1/k^2$ tail, complement $1-1/k^2$	the whole topic
H3	memoryless $P(T>s+t T>s)=P(T>t)$, $e^{-1} = 0.3679$	exponential
H4	68.27 / 95.45 / 99.73	normal landmarks
H5	expected count = $N \times P$	the count question
H6	geometric $E = 1/p$	discrete lifeline
H7	hypergeometric mean = nK/N	without replacement
H8	$\phi(z)$ area-from-0 vs $F(z)$ cumulative	the table decision

Study order for these: H2 and H1 first (they gate the most), then H8 and H3, then H5, then H6, H7, H4. Total 2h, and it is the best-spent time before the paper.